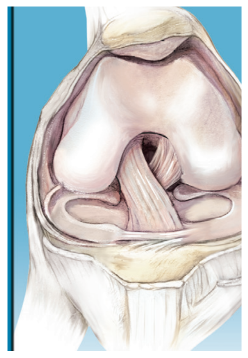

93

Arthroscopic Synovectomy of the Knee

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The synovial lining is a specialized mesenchymal tissue that is integral to the normal functioning of a joint. Synovial disorders can involve varying amounts of the synovium. Rheumatoid arthritis shows total joint involvement, whereas on the other end of the spectrum, plica syndrome is caused by an isolated synovial lesion.

Volkman performed the first synovectomy in 1855 for tuberculous synovitis. Although the indications and technique have changed over time, the procedure is still performed, and the objective of removing the diseased synovium remains the same.[1] Compared with open procedures, arthroscopic techniques have enabled surgeons to perform a synovectomy without a large arthrotomy, decreasing the risk of postoperative arthrofibrosis. Arthroscopy also serves as an effective technique to remove synovium in the posterior compartment and allows viewing of synovial lesions that may be missed with open procedures. Arthroscopic synovectomy can be used in the surgical treatment of rheumatoid arthritis, pigmented villonodular synovitis (PVNS), hemophilic synovitis, plicae, synovial hemangioma, synovial osteochondromatosis, and degenerative synovitis.

As with all orthopedic conditions, a complete workup including a thorough history and physical examination and complete imaging analysis is needed to evaluate these patients. In addition, a trial of medical management should be performed before initiation of surgical treatment. Surgical treatment consists of arthroscopically removing varying amounts of synovium, the amount of which is based on the underlying disease process.

History

A complete history is important in the evaluation of patients with synovial disorders. The presence of other affected joints, chronicity of symptoms, exacerbating factors, and the amount of disability experienced by the patient on a daily basis are important pieces of information. Patients with rheumatoid arthritis may have more systemic complaints, including morning stiffness and other affected joints, particularly the small joints of the hands and feet. PVNS is typically a monoarticular process that affects adults in the third or fourth decade of life. Symptoms are mechanical in nature and may be similar to those seen in patients with meniscal tears.[2] Clinically patients have the insidious onset of localized warmth, swelling, and stiffness with occasional locking and a palpable mass. Plica syndrome is a finding in patients with anteromedial knee pain. Patients experience tightness, snapping, giving way, and pain with repetitive activities. Clinically it is difficult to distinguish plica syndrome from other causes of knee pain such as meniscal tears, patellar tendinitis, or patellofemoral pain syndrome.

Physical Examination

Other joints may be affected in patients with rheumatic or autoimmune disorders, and these joints should be evaluated. Patients with rheumatoid arthritis often have a flexion contracture and quadriceps atrophy in the knee region.[3] The skin should be examined, as well as previous incisions and subcutaneous nodules. Knee examination includes overall alignment, range of motion (ROM), the presence of an effusion, warmth, tenderness, crepitus, strength, meniscal integrity, and stability. Collateral ligament instability or bony malalignment suggests more severe articular loss, and patients with these conditions are poor candidates for a synovectomy.

The physical examination for PVNS is often nonspecific. An effusion is associated with diffuse involvement. Palpation of the joint may show warmth and tenderness. Aspiration of the joint fluid may show a dark-brown fluid that is a result of recurrent bleeding into the joint. Cytologic studies of the aspirate may show hemosiderin pigment and multinucleated foreign body giant cells, but often the findings of these studies are normal.[4] Ligamentous instability is uncommon in persons with PVNS.

Plica syndrome begins insidiously. Tenderness over the medial parapatellar region is common. A plica may sometimes be directly palpated and rolled under the finger, recreating the patient's symptoms. If the medial border of the patella is palpated while pushing the patella medially with one hand and the other hand produces a valgus stress with external rotation of the tibia, pain may be elicited, suggesting plica syndrome.[5] An effusion is not typically present in persons with plica syndrome.

Imaging

Patients with rheumatoid arthritis may have periarticular erosions and osteopenia. Cervical spine flexion and extension views should be obtained in preoperative patients to rule out cervical instability. Radiographs in patients with PVNS can show erosive, cystic, and sclerotic lesions of the articular surface. If enough synovium that contains hemosiderin is present, soft tissue masses may be seen, but often the findings of the films are normal with well-maintained joint spaces. Magnetic resonance imaging is considered to be the most diagnostic study for PVNS. It may show nodular intra-articular masses of low signal intensity on

Abstract

Arthroscopic synovectomy is a treatment option for a number of synovial disorders, including rheumatoid arthritis, pigmented villonodular synovitis, hemophilic synovitis, plicae, synovial hemangioma, synovial osteochondromatosis, and degenerative synovitis. The procedure requires thoughtful preparation and extensive preoperative workup. Surgical approach utilizes multiple portals to allow access to all areas of the knee joint to remove diseased synovium. Results vary based on the underlying etiology, but the procedure does appear to help patients with the symptomatic management of synovial disorders. Complications are rare but include recurrent hemarthrosis, loss of range of motion, neurovascular injury, and infection.

Keywords

knee arthroscopy
synovectomy
pigmented villonodular synovitis
rheumatoid arthritis
radiation

T1- and T2-weighted images and also allows for the evaluation of the location and extent of disease.

DECISION-MAKING PRINCIPLES

A full rheumatologic workup should be completed for patients with systemic diseases, and appropriate laboratory tests should be up to date. Patients with hemophilia require a consultation with a hematologist. If surgical treatment is to be pursued, it is essential to have a well-thought-out plan for perioperative management of clotting factors.

In disorders associated with a localized lesion, such as a localized PVNS (Figs. 93.1 and 93.2) or plica, arthroscopic intervention can remove the pathology in its entirety. Persons with diffuse conditions such as rheumatoid arthritis (Figs. 93.3 through 93.6) or hemophilia can undergo surgery to decrease the severity of disease symptoms once conservative measures have been exhausted.

Treatment Options

Treatment options vary based on etiology. Recently medical management of rheumatoid arthritis has improved significantly. The goals of medical treatment include reducing the number of painful and swollen joints, suppressing the acute phase response, decreasing the rheumatoid factor titer, and slowing radiographic progression of the disease. Medical management should consist of a combination of disease-modifying antirheumatic drugs, nonsteroidal antiinflammatory drugs, an appropriate physical therapy regimen, activity modification, and intra-articular steroid injections.[6] A patient with rheumatoid arthritis and minimal degenerative changes on radiographs would be a candidate for arthroscopic synovectomy after the failure of approximately 6

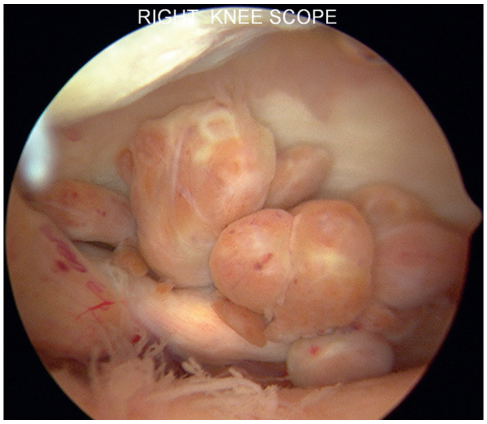

Fig. 93.1 Arthroscopic appearance of localized pigmented villonodular synovitis.

Fig. 93.2 Gross view of the localized pigmented villonodular synovitis specimen from Fig. 93.1.

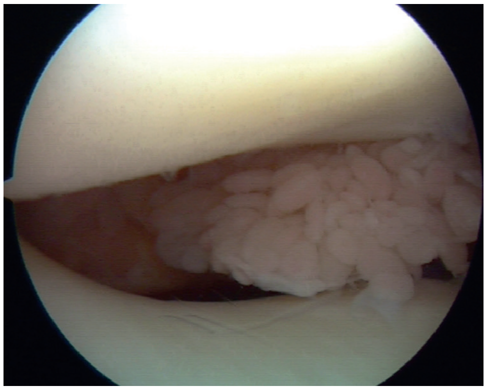

Fig. 93.3 The patellofemoral joint of a left knee in a 39-year-old patient with rheumatoid arthritis. Despite conservative treatment, the patient experienced a flexion contracture with uncontrolled swelling and elected to undergo arthroscopic synovectomy.

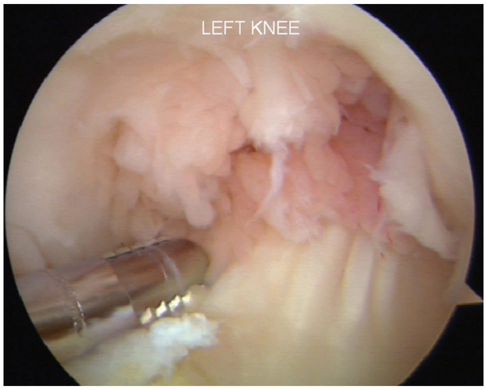

Fig. 93.4 The arthroscopic view of the notch before débridement in the left knee of the patient in Fig. 93.3.

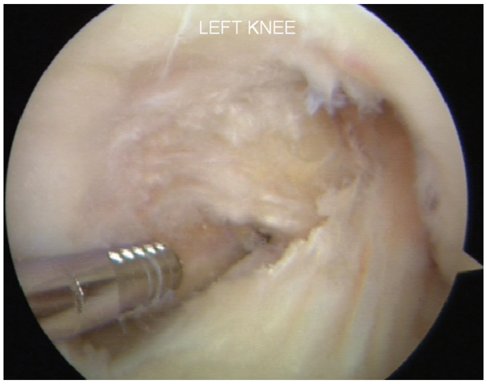

Fig. 93.5 The arthroscopic view of the notch after débridement in the same patient shown in Fig. 93.3.

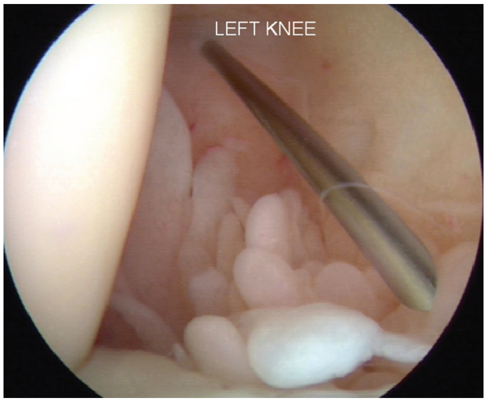

Fig. 93.6 The arthroscopic view of the posteromedial compartment of the patient shown in Fig. 93.1. The spinal needle is used to localize the ideal portal location.

months of medical management. Significant joint space narrowing or mechanical malalignment is a relative contraindication to synovectomy for inflammatory synovial knee disorders, and a total knee arthroplasty is recommended.

Hemophilic synovitis can also be associated with significant joint destruction and has shown favorable improvement in symptoms with synovectomy.[7–9] Radiosynovectomy is indicated as the first procedure in persons with hemophilic synovitis, with satisfactory results in 80% of patients.[10] No more than three radiosynovectomies can be performed per year. If the three radiosynovectomy procedures fail to relieve symptoms, an arthroscopic synovectomy is indicated.[10] Although joint deterioration is not preventable, a synovectomy can reduce recurrent hemarthrosis and maintain ROM. A hemophilic synovectomy requires an inpatient stay for coordinated management of clotting factors with the patient's hematologist.

In PVNS, arthroscopic synovectomy is the treatment of choice for localized disease, but open synovectomy may be required for diffuse-type PVNS and recurrence of local tumors. Adjuvant therapies including intra-articular radiotherapy or moderate-dose external beam radiotherapy further reduce likelihood of recurrence with advanced forms and recurrent disease.[11]

Authors' Preferred Technique

Arthroscopic synovectomy requires the use of multiple portals to access all spaces in the knee joint; therefore detailed preoperative planning and patient setup is essential for a successful operation. General anesthesia is recommended, and the use of a Foley catheter should be considered. An epidural can be used if required for medical reasons and may also help with postoperative pain relief.

Examination After Inducement of Anesthesia

Both knees are examined for an assessment of the ROM, ligamentous stability, patellar mobility, patellar tracking, and the presence of an effusion.

Positioning

The patient is placed supine on the operating room table. The well leg is appropriately padded and secured in a well leg holder after placement of a compressive stocking and sequential compression device. A leg holder is not used on the operative leg because it may interfere with the use of the superomedial and superolateral portals. The foot of the operating table is dropped and the mid portion of the table is flexed to avoid hip hyperextension. A well-padded thigh tourniquet is placed high on the operative leg.

Surgical Steps

Anterior Compartments

After standard prepping and draping is performed, the extremity is exsanguinated and the tourniquet is inflated to 300 mm Hg. The tissue obtained from the synovectomy should be collected and sent for pathologic evaluation. The anterior aspect of the knee is addressed first. A superomedial outflow portal is created, and the outflow cannula is placed here. Standard inferolateral and inferomedial portals are created. The arthroscope is placed in the inferolateral portal, and an initial diagnostic arthroscopy is performed. The synovectomy then proceeds with the use of an arthroscopic shaver. While viewing from the inferolateral portal with the knee in extension, the shaver is used in the superolateral and inferomedial portals to remove all synovial tissue, but avoiding injury to surrounding muscle, tendon, and fascia (Table 93.1).

Posterior Compartments

The anterior compartment should now be finished, and attention is turned to the posterior compartments, beginning with the posteromedial compartment. Typically the posterior compartments can be visualized with a 30-degree scope; however, a 70-degree scope can be used if difficulty is encountered. The arthroscope must first be placed in the posterior compartment. The blunt-tipped trocar is placed in the arthroscopic sheath and inserted into the inferolateral portal. The trocar is directed toward the medial femoral condyle, and when it is contacted, the trocar is carefully advanced posteriorly through the interval between the medial femoral condyle and the posterior cruciate ligament, raising the hand with insertion to match the slope of the tibia. If this maneuver proves difficult, a central and vertically oriented patellar tendon portal may provide easier access to the posterior compartment. The arthroscope is inserted, and the posterior portion of the medial femoral condyle and the posterior horn of the medial meniscus should be visible (see Fig. 93.6). While looking medially, a spinal needle is inserted anterior to the medial head of the gastrocnemius into the posteromedial compartment. The needle is used to ensure that all areas in the posterior knee that are in need of synovectomy can easily be reached. Once the ideal portal location has been determined, a longitudinal incision is made in the skin. A hemostat is used to bluntly dissect and then penetrate the capsule, and a cannula is placed. The shaver is placed through the cannula, and the synovium in the posteromedial compartment is resected.

The posterolateral compartment is accessed in a manner similar to the posteromedial compartment. A blunt-tipped trocar is placed in the arthroscopic cannula in the inferomedial portal between the lateral femoral condyle and the anterior cruciate ligament. The hand is gently raised and advanced posteriorly,

TABLE 93.1 Steps for Performing an Arthroscopic Synovectomy of the Knee

Step	Area of Synovial Resection	Camera Portal/Leg Position	Instrument Portal
1	A. Suprapatellar pouch B. Lateral gutter	Inferolateral/extension	Superolateral
2	A. Begin medial gutter B. Medial aspect suprapatellar pouch C. Intercondylar notch	Inferolateral/extension	Inferomedial
3	A. Finish medial gutter B. Medial suprapatellar pouch	Inferolateral/flexion	Superomedial
4	A. Retropatellar space B. Inferolateral gutter	Superolateral/extension	Inferolateral
5	A. Retropatellar space B. Inferomedial gutter	Superolateral/extension	Inferomedial
6	A. Posteromedial compartment	Inferolateral/flexion	Posteromedial
7	A. Posterolateral compartment	Inferomedial/flexion	Posterolateral

taking care not to violate the posterior capsule, which could put the neurovascular structures at risk. The arthroscope replaces the trocar, and the posterior lateral femoral condyle and the posterior horn of the lateral meniscus are viewed. Again, a spinal needle is used to make a posterolateral portal under direct visualization. Placing the needle in the soft spot, anterior to the biceps femoris muscle and posterior to the iliotibial band, helps protect the common peroneal nerve. The needle should be inserted posterior to the fibular collateral ligament and anterior to the lateral head of the gastrocnemius. Once it is determined that the spinal needle is placed so that all areas that require synovectomy can be reached, the skin is incised and a hemostat is used to dissect to the posterior capsule. The capsule is then punctured under direct visualization and a cannula is placed. The shaver is placed through the cannula, and the posterolateral compartment synovectomy is performed.

After the synovectomy is complete, the tourniquet is deflated and an electrocautery device is used to achieve hemostasis. It is common to use a suction drain for 24 hours to help minimize hemarthrosis. Ice, elevation, and a light compressive dressing are used to minimize swelling, and early motion is encouraged.

Postoperative Management

The patient can bear weight as tolerated after surgery. Physical therapy is begun on postoperative day 1 after drain removal and concentrates on closed chain exercises. The most immediate goals are regaining knee extension and quadriceps function. A continuous passive motion machine can be used for the first few days after surgery.

RESULTS

Goetz et al.[12] evaluated 32 knees at 14-year follow-up after combined arthroscopic and radiation synovectomy for rheumatoid arthritis. These investigators concluded that combined arthroscopic and radiation synovectomy led to a stable improvement of knee function for a minimum of 5 years, but repeat surgery was frequent, with 56% of patients having another operation by 10 years.[12] If total knee arthroplasty was considered the end point, the joint survival rate was 88.5% at 5 years, 53.9% at 10 years, and 39.6% at 14 years.[12] Carl et al.[13] studied 11 patients with rheumatoid arthritis who were undergoing a synovectomy. These investigators found that a synovectomy led to an overall reduction of acute inflammatory infiltrates by 82.1% and of chronic inflammatory infiltrates by 62.5%.[13]

A meta-analysis of 630 patients with PVNS who underwent arthroscopic, combined anterior arthroscopic, and posterior open synovectomy, or open synovectomy had an overall recurrence rate of 21% (137/630).[11] A multivariate analysis of recurrence showed no difference between surgical approaches for localized disease, and low-quality evidence of significant reduction by open (OR = 0.47; 95% CI 0.25 to 0.90; P = .024) or combined (OR = 0.19; 95% CI 0.06 to 0.58; P = .003) synovectomy versus with arthroscopic surgery. Rates of wound complications and reoperation were similar between groups, but the open synovectomy group had significantly increased postoperative stiffness (10.5%, 37/354) versus the combined group (2.7%, 1/37) and the arthroscopic synovectomy group (2.1%, 5/239).[11]

Although few follow-up data are available regarding synovectomy in patients with hemophilia in general, Verma et al.[14] suggest that the primary predictor of outcome in hemophiliacs is the degree of intra-articular preexisting degenerative changes. In more severe cases, the results of arthroscopic synovectomy are less predictable, and total joint arthroplasty should be considered.

Complications

After synovectomy, a recurrent hemarthrosis may occur. The fluid can often be aspirated with a large-bore needle, but an arthroscopic washout may be needed. Joint stiffness and loss of extension are also possible. Performing aggressive and early ROM exercises, using extension boards, and dynamic bracing may help with these symptoms. A septic joint or neurovascular injury can also occur. With use of careful technique, these complications can be minimized.

FUTURE DIRECTIONS

Future directions include adjuvant therapies to reduce likelihood of recurrence of disease following arthroscopic synovectomy. For example, PVNS tumors are stimulated by the overexpression of macrophage colony-stimulating factor (M-CSF) by synovial fibroblasts. Tyrosine kinase inhibitors, such as imatinib, have been shown to reduce tumor progression and may become an effective treatment for advanced form and recurrent disease.[15]

For a complete list of references, go to ExpertConsult.com

SELECTED READINGS

Citation:
Goetz M, et al. Combined arthroscopic and radiation synovectomy of the knee joint in rheumatoid arthritis: 14-year follow-up. *Arthroscopy*. 2011;27(1):52–59.

Level of Evidence:
IV, therapeutic case series

Summary:
The authors evaluated the outcome of 32 knees treated with combined arthroscopic and radiation synovectomy of the knee joint in early cases of rheumatoid arthritis in terms of knee function and the need for repeat surgical intervention. At an average of 14 years of follow-up, with any repeat surgical intervention as the end point, the survival rate was 32%, challenging the long-term benefit of the procedure.

Citation:
Carl HD, et al. Site-specific intraoperative efficacy of arthroscopic knee joint synovectomy in rheumatoid arthritis. *Arthroscopy*. 2005;21(10):1209–1218.

Level of Evidence:
III

Summary:
The authors assessed site-specific intraoperative reduction of inflammatory infiltrates achieved with arthroscopic knee synovectomy in patients with rheumatoid arthritis using preoperative and postoperative synovial tissue samples. They concluded that arthroscopic synovectomy reduces acute and chronic inflammatory infiltrates in patients with rheumatoid arthritis. However, this reduction appears to depend on the anatomic region of the joint.

Citation:
Mollon B, Lee A, Busse JW, et al. The effect of surgical synovectomy and radiotherapy on the rate of recurrence of pigmented villonodular synovitis of the knee. *Bone Joint J*. 2015;97-B(4): 550–557.

Level of Evidence:
IV, therapeutic study

Summary:
The authors performed a meta-analysis of 630 patients who underwent surgical synovectomy of the knee for PVNS. The overall rate of recurrence was 21.8% (137/630). Open synovectomy and combined arthroscopic/open synovectomy had lower recurrence rate than arthroscopic synovectomy for diffuse PVNS, but not for localized disease. Perioperative radiotherapy reduced recurrence of diffuse PNVS. Complication rates were similar except for significantly increased post-operative stiffness in the open and combine arthroscopic/open synovectomy groups.

Citation:
Kim TK, et al. Neurovascular complications of knee arthroscopy. *Am J Sports Med*. 2002;30:619–629.

Level of Evidence:
IV, review article

Summary:
The authors summarize causes and frequencies of complications related to arthroscopic procedures, including anterior and posterior compartment synovectomy. Complication management and medicolegal implications are reviewed.

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